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(54) **POSITIONING AND CARRYING STRUCTURE, AUTOMATED GUIDED VEHICLE AND LOADING SYSTEM**

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***B66F 9/18*** (2006.01)
***B66F 9/19*** (2006.01)

(52) **U.S. Cl.**
CPC ................ ***B66F 9/195*** (2013.01); ***B66F 9/18*** (2013.01)

(58) **Field of Classification Search**
CPC ...... H01L 21/67724; B66F 9/183; B66F 9/18; B66F 9/195
See application file for complete search history.

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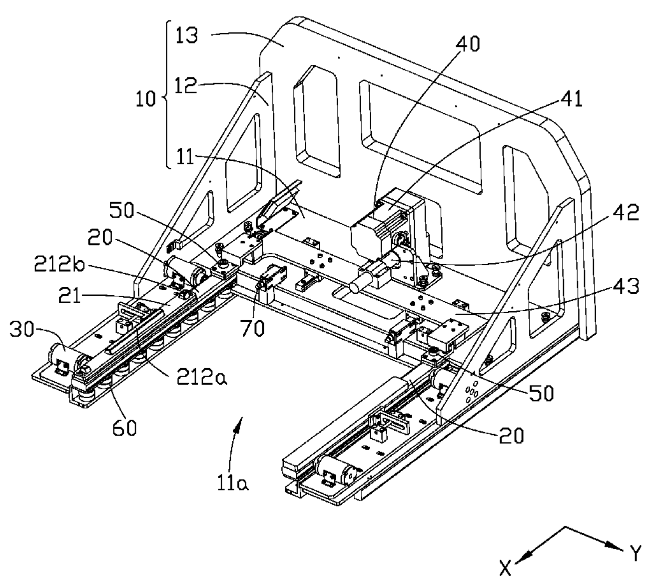
(57) **ABSTRACT**

A carrier applying precision support and lifting by an Automated Guided Vehicle includes a frame, two supporting arms, and pressing components. Two supporting arms are located on either side of the frame and are movable in a receiving direction relative to the frame. The pressing components are located on the frame, the pressing components include first pressure balls, for pushing the supporting arms across the carrier, and second pressure balls for pushing the supporting arms along the receiving direction. The first and second balls improve the positioning accuracy of the supporting arms relative to the object or material to be transported and reduces friction between the supporting arms and the pressing components. An AGV cart and a loading system are also disclosed.

**19 Claims, 5 Drawing Sheets**

100



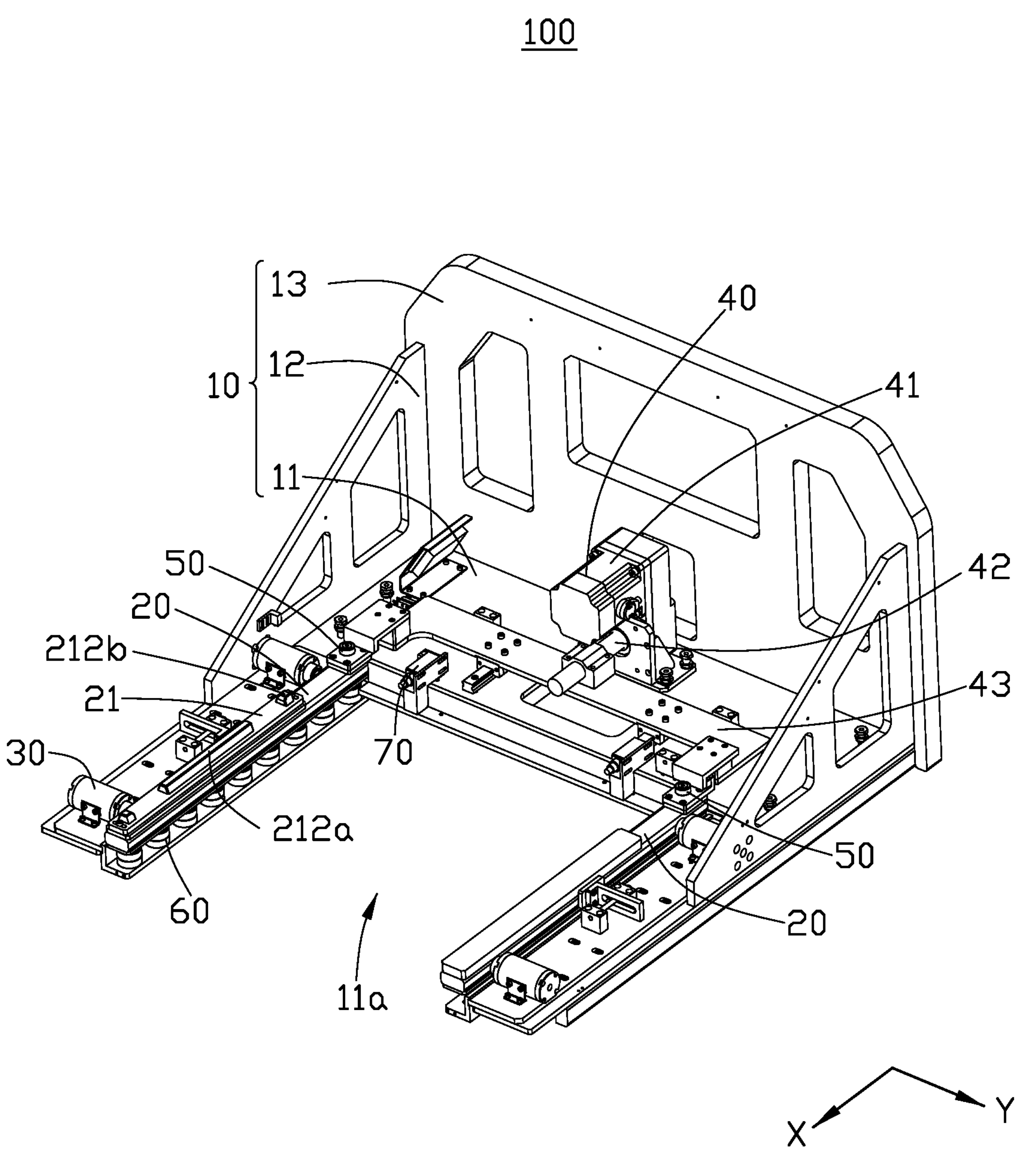



FIG. 1

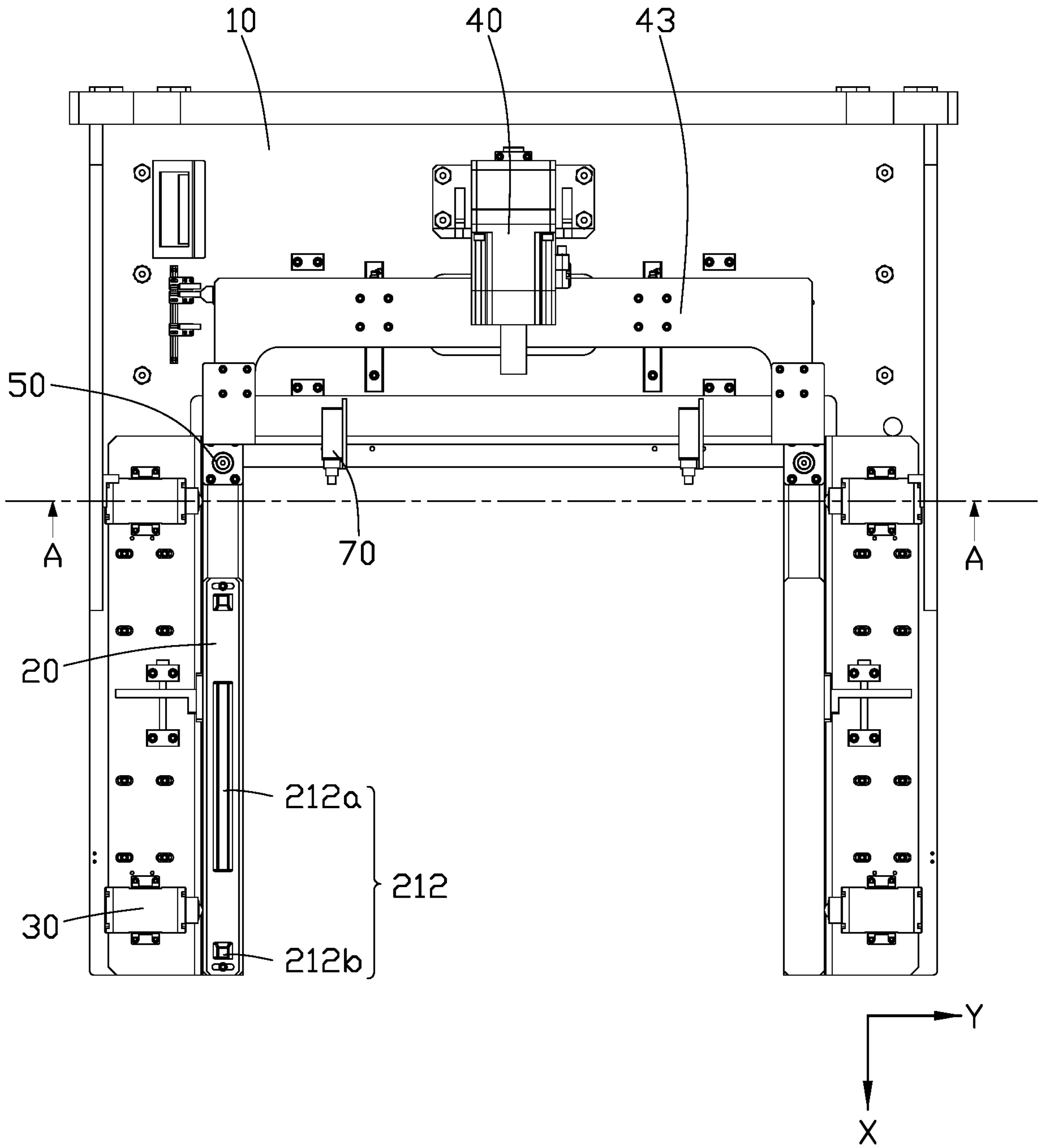



FIG. 2

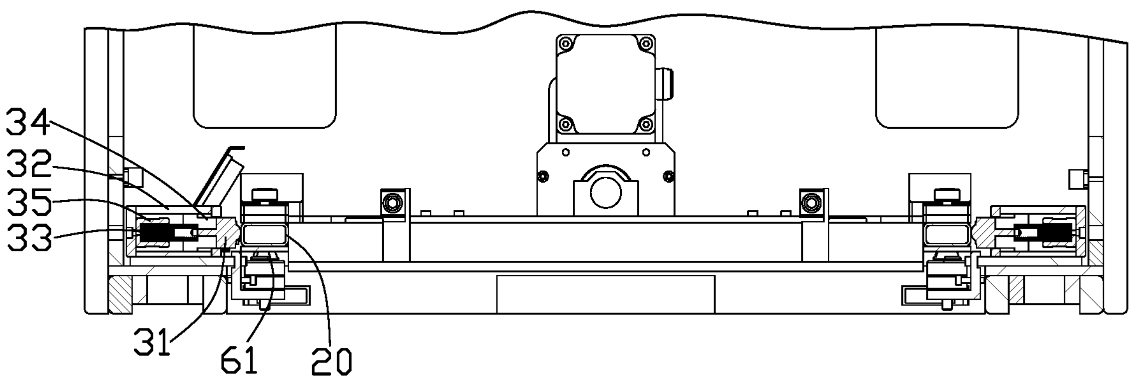



FIG. 3

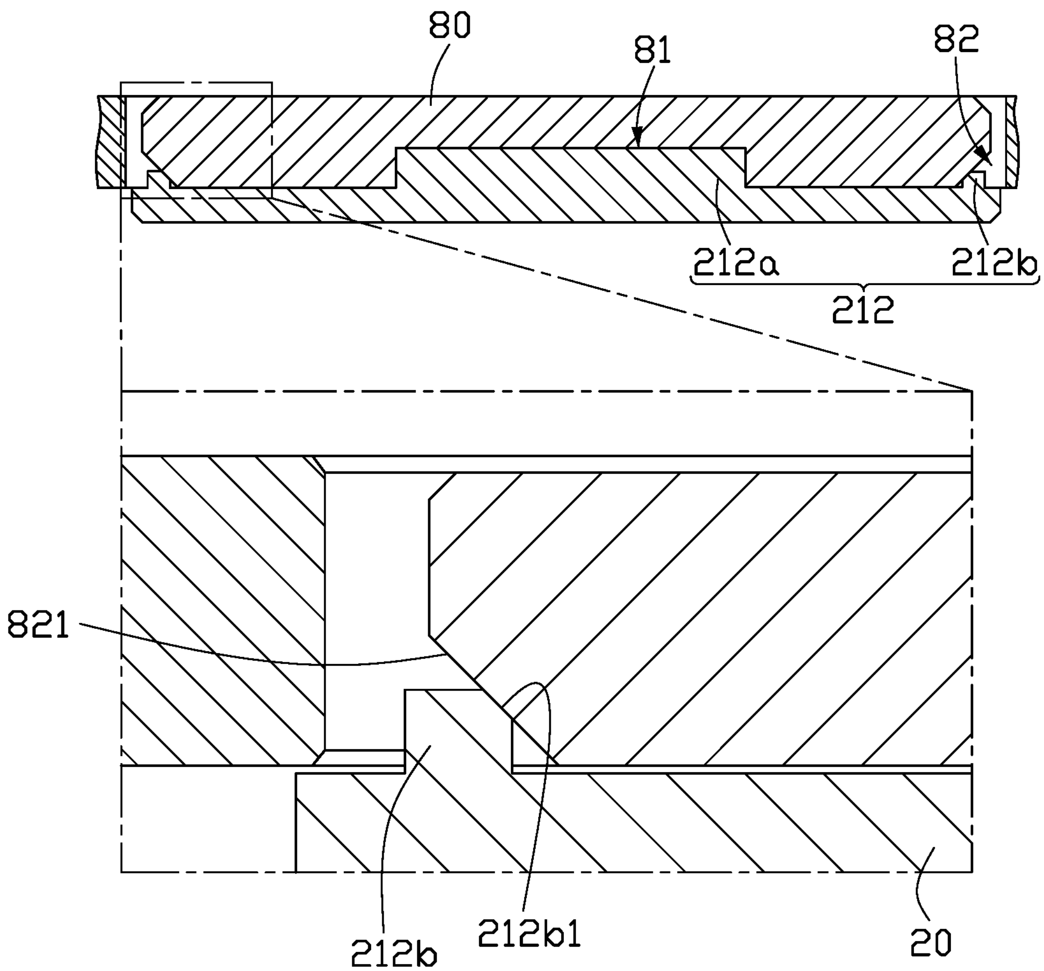



FIG. 4

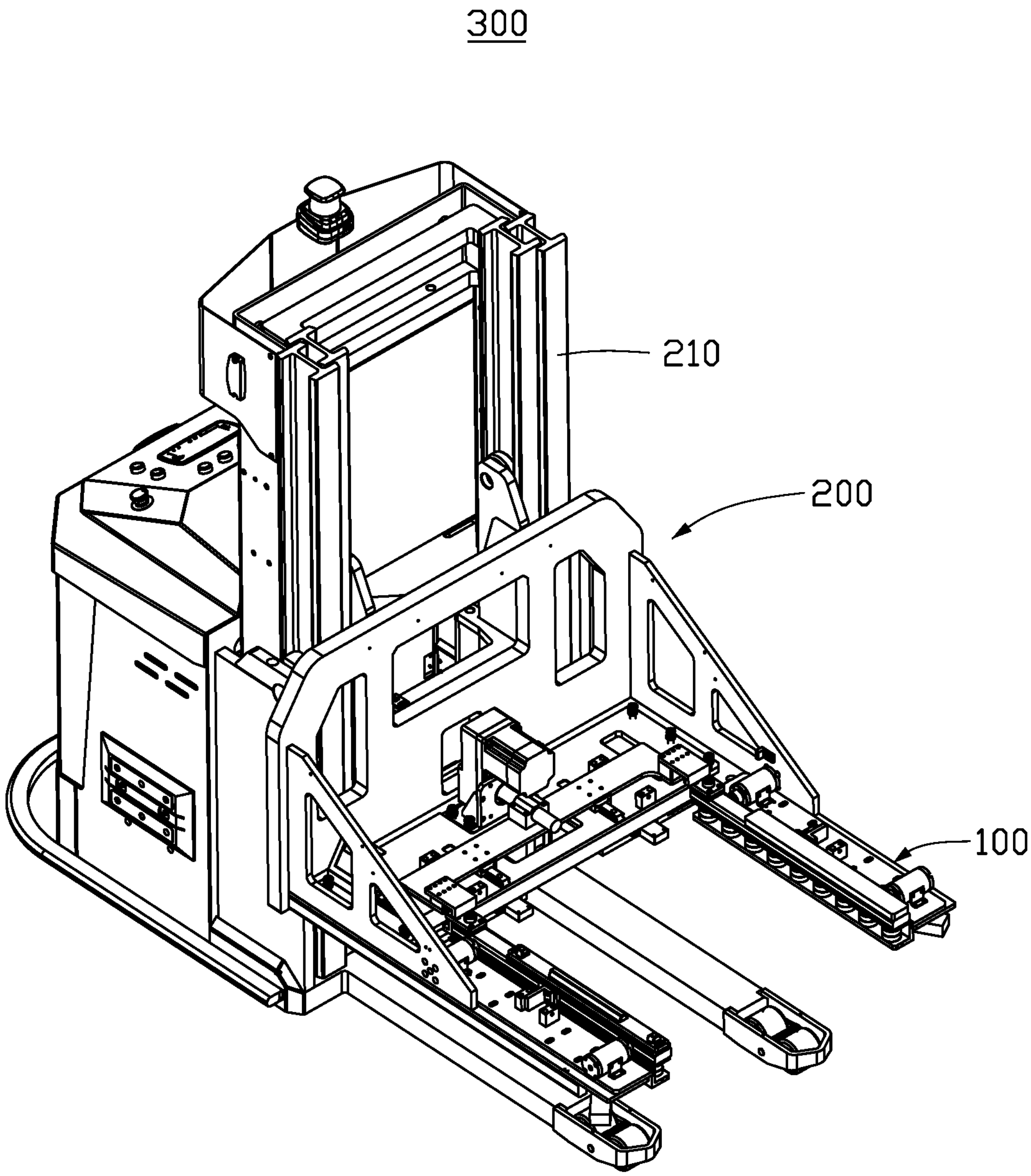



FIG. 5

# POSITIONING AND CARRYING STRUCTURE, AUTOMATED GUIDED VEHICLE AND LOADING SYSTEM

## FIELD

The subject matter herein generally relates to materials handling, and to carrier, AGV cart and loading system.

## BACKGROUND

In industrial production, the AGV cart commonly carries materials and delicate objects. However, the position of the materials may be different each time, resulting in low positioning accuracy and thus a risk of the materials being shed or damaged.

## BRIEF DESCRIPTION OF THE DRAWINGS

Many aspects of the disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

FIG. **1** is an isometric view of a materials carrier according to an embodiment of the present disclosure.

FIG. **2** is a top view of the carrier according to an embodiment of the present disclosure.

FIG. **3** is a front view of the carrier according to an embodiment of the present disclosure.

FIG. **4** is a cross-sectional view along line A-A in FIG. **2**.

FIG. **5** is an isometric view of an Automated Guided Vehicle cart according to an embodiment of the present disclosure.

## DETAILED DESCRIPTION

It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale and the proportions of certain parts have been exaggerated to better illustrate details and features of the present disclosure.

The present disclosure, including the accompanying drawings, is illustrated by way of examples and not by way of limitation. Several definitions that apply throughout this disclosure will now be presented. It should be noted that references to "an" or "one" embodiment in this disclosure are not necessarily to the same embodiment, and such references mean "at least one."

The term "coupled" is defined as connected, whether directly or indirectly through intervening components, and is not necessarily limited to physical connections. The connection can be such that the objects are permanently connected or releasably connected. The term "comprising" means "including, but not necessarily limited to"; it specifically indicates open-ended inclusion or membership in a so-described combination, group, series, and the like.



Without a given definition otherwise, all terms used have the same meaning as commonly understood by those skilled in the art. The terms used herein in the description of the present disclosure are for the purpose of describing specific embodiments only, and are not intended to limit the present disclosure.

As shown in FIG. **1** to FIG. **3**, a structure (carrier **100**) to take a precision hold of and transport industrial materials on an Automated Guided Vehicle (AGV) of an embodiment includes a frame **10**, at least two supporting arms **20** and at least one press component **30**. The frame **10** includes horizontal board **11**, stiffener **12**, and vertical board **13**. The horizontal board **11** is provided with an opening **11***a*. The opening **11***a* is configured for containing an object to be transported. The stiffener **12** is configured for improving the overall strength of the connection of horizontal board **11** and vertical board **13**. The supporting arms **20** and the press component **30** are installed on the horizontal board **11**. The vertical board **13** is configured for installing the carrier **100**.

As shown in FIG. **1** to FIG. **3**, the two supporting arms **20** are extended along direction X, and arranged side by side in direction Y. The direction Y is perpendicular to the direction X. The two supporting arms **20** are movable relative to the horizontal board **11** along the horizontal direction X. The two supporting arms **20** are on either side of the opening **11***a*. The two supporting arms **20** support and hold an object together.

The carrier **100** also includes a drive mechanism **40**. The drive mechanism **40** is configured for driving the supporting arms **20** to move along direction X. The drive mechanism **40** includes a motor **41**, a transmission assembly **42**, and a moving arm **43**. The motor **41** and the transmission assembly **42** are mounted on the horizontal board **11**. The moving arm **43** is movable relative to the frame **10** along the direction X. The transmission assembly **42** is connected to the motor **41** and the moving arm **43**, and converts the rotation of the motor **41** into movement of the moving arm **43**. The moving arm **43** is connected to the two supporting arms **20**, and the supporting arms **20** and moving arm **43** can be moved simultaneously. As an example, the distance of movement of the supporting arms **20** is 0-30 mm.

As shown in FIG. **1** to FIG. **3**, in some embodiments, the transmission assembly **42** can be a timing belt, a gear, etc., and the output end of the transmission assembly **42** is a screw, which moves the moving arm **43** along the direction X.

In some embodiments, a gap is defined between the moving arm **43** and each supporting arm **20** in the direction X to allow positional adjustment of the supporting arm **20** relative to the moving arm **43**. An adjusting bolt **50** connects the moving arm **43** and the supporting arm **20**, to fix the position of the supporting arm **20** relative to the moving arm **43**. The adjusting bolt **50** can be made tighter or looser manually. When the adjusting bolt **50** is loosened, the position of the unlocked supporting arm **20** can be adjusted slightly. In a preferred position of the supporting arm **20**, the adjusting bolt **50** can be tightened to fix and lock the supporting arm **20** in that position, which improves the positioning accuracy of the supporting arm **20** relative to the object to be transported.

In some embodiments, a positioning plate **21** is located on each supporting arm **20**. The positioning plate **21** can move along the direction Y. The positioning plate **21** is provided with at least one positioning column **212**. The positioning column **212** is configured for positioning the object to be transported, so as to improve the stability of the material.

As shown in FIG. **4**, in some embodiments, the object for transportation is a fixture **80**. The fixture **80** is a structure configured for fixing a workpiece or material stably in a certain orientation. The fixture **80** is provided with at least one positioning groove on both sides. There may be two kinds of positioning column **212**, one a long column **212***a*, the other a short column **212***b*. One long column **212***a* and two short columns **212***b* are mounted on each supporting arm **20** and arranged along the direction X. The long column **212***a* is located between the two short columns **212***b*. There may also be two kinds of positioning groove, one a long groove **81**, the other a short groove **82**. Each side of the fixture **80** along the direction Y each is provided with one long groove **81** and two short grooves **82**. The long groove **81** is between the two short grooves **82**. The positions of the long column **212***a* and the long groove **81** correspond one-to-one, and the positions of the short columns **212***b* and the short grooves **82** also correspond one-to-one. The long column **212***a* can be inserted into the long groove **81**, and the short columns **212***b* can be inserted into the short groove **82**.

In some embodiments, the two short grooves **82** are located at opposite ends of one side of the fixture **80**. A first slope **821** is formed in each short groove **82**. Each short column **212***b* is provided with a second slope **212***b***1** going upwards. The inclination of the first slope **821** is equal to the inclination of the second slope **212***b***1**. When the short column **212***b* is inserted into the short groove **82**, the second slope **212***b***1** rests against the first slope **821**, limiting the position of the fixture **80**.

As shown in FIG. **1** and FIG. **3**, the carrier **100** includes four press components **30**. Two press component **30** are located on each supporting arm **20**, each being close to an end of the arm **20**. Each press component **30** includes a first ball **31**. The first ball **31** can push the supporting arm **20** along the direction Y, to improve the positioning accuracy between the supporting arm **20** and the fixture **80** and the consequent stability of transportation, and friction between the press component **30** and the supporting arm **20** is also reduced.

In some embodiments, each press component **30** includes a shell **32**, an elastic member **33**, and a guidepost **34**. The elastic member **33** and the guidepost **34** are accommodated in the shell **32**. One end of the elastic member **33** rests against the inner wall of the shell **32**, and the other end bears against the guidepost **34**. The end of the guidepost **34** away from the elastic member **33** is connected with the first ball **31**, which can move the shell **32** along the direction Y. As an example, the range of movement of the first ball **31** is 0-15 mm.

In some embodiments, each press component **30** also includes a spring cover **35**. The elastic member **33** is a spring. The spring cover **35** is accommodated in the shell **32**. One end of the elastic member **33** is held in the spring cover **35**, for stability of the elastic member **33**.

In some embodiments, the frame **10** is provided with four adjusting grooves, each shell **32** is inserted in one adjusting groove to allow the press component **30** to change position relative to the frame **10** along the direction Y.

In some embodiments, the carrier **100** also includes a plurality of support components **60**. Support components **60** are arranged on the frame **10** along the direction X below each supporting arm **20**. Support components **60** are configured for supporting the supporting arm **20**. Each support component **60** includes a second ball **61**. The second ball **61** is located below the supporting arm **20** and supports the bottom of the supporting arm **20**. A plurality of second balls **61** can improve the stability of the supporting arm **20** and reduces friction between the supporting arm **20** and the support components **60**.



In some embodiments, a limit switch **70** is located on the frame **10** between the two supporting arms **20**. The limit switch **70** can sense the object to be carried, so as to improve the positioning accuracy.

As shown in FIG. **5**, an automated guided vehicle (AGV cart **300**) of an embodiment includes a car **200** and the carrier **100**. The car **200** includes a lift **210**. The carrier **100** is mounted on the lift **210**. The lift **210** can move the carrier **100** up or down.

A loading system of an embodiment includes the AGV cart **300** and the fixture **80**. To unload the fixture **80**, the car **200** moves to the fixture **80** until the supporting arms **20** is below the fixture **80**, and the lift **210** lifts the carrier **100** to bring the supporting arms **20** close to the fixture **80**. In the meantime, the drive mechanism **40** drives the supporting arms **20** move along direction X until the positioning columns **212** are aligned with the positioning grooves. After the positioning columns **212** are inserted into the positioning grooves, the lift **210** lifts the carrier **100** with the fixture **80** and the car **200** can move to its destination.

The embodiments shown and described above are only examples. Even though numerous characteristics and advantages of the present technology have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the disclosure is illustrative only, and changes may be made in the detail, including in matters of shape, size and arrangement of the parts within the principles of the present disclosure up to, and including the full extent established by the broad general meaning of the terms used in the claims.

What is claimed is:

**1**. A positioning and carrying structure comprising:
- a frame;
- at least two supporting arms located on the frame and movable relative to the frame along a direction X; and
- a plurality of press components located on the frame, wherein
- each of the plurality of press components comprises:
- a first ball located on a side of a supporting arm of the at least two supporting arms away from another supporting arm of the at least two supporting arms, and the first ball configured to push the supporting arm elastically along a direction Y, wherein
- the direction Y is substantially perpendicular to the direction X, the plurality of press components is arranged along the direction X on the side of each of the at least two supporting arms.

**2**. The positioning and carrying structure of claim **1**, further comprising a drive mechanism, wherein:
- the drive mechanism comprises a motor, a transmission assembly and a moving arm, the transmission assembly is connected to the motor and the moving arm, the moving arm is located on the frame and movable relative to the frame along the direction X, the moving arm is connected to the at least two supporting arms, the at least two supporting arms move with the moving arm.

**3**. The positioning and carrying structure of claim **2**, wherein:
- a gap is defined between the moving arm and the each of the at least two supporting arms in the direction X, the each of the at least two supporting arms is configured to change a position relative to the moving arm;

an adjusting bolt is connected between the moving arm and the each of the at least two supporting arms to fix a position of the each of the at least two supporting arms relative to the moving arm.

**4**. The positioning and carrying structure of claim **1**, further comprising a positioning plate, wherein:

the positioning plate is located on the each of the at least two supporting arms the positioning plate is provided with a positioning column, the positioning column is configured for positioning an object to be transported.

**5**. The positioning and carrying structure of claim **1**, further comprising:

a plurality of support components arranged on the frame along the direction X below each of the at least two supporting arms, wherein:

each of the plurality of support components comprises:

a second ball located below the each of the at least two supporting arms to support the each of the at least two supporting arms.

**6**. The positioning and carrying structure of claim **1**, wherein:

each of the plurality of press components comprises:

a shell;

an elastic member accommodated in the shell;

a guidepost accommodated in the shell;

one end of the elastic member rests against the inner wall of the shell, and another end of the elastic member bears against the guidepost, an end of the guidepost away from the elastic member is connected with the first ball.

**7**. The positioning and carrying structure of claim **6**, wherein:

the frame comprises at least one adjusting groove, the shell is inserted in one adjusting groove of the at least one adjusting groove, a corresponding press component of the plurality of press components is configured change a position relative to the frame by adjusting the shell within the adjusting groove.

**8**. The positioning and carrying structure of claim **1**, further comprising:

a limit switch located on the frame between the at least two supporting arms.

**9**. An automated guided vehicle comprising:

a car;

a positioning and carrying structure located on the car, wherein

the positioning and carrying structure comprises:

a frame;

at least two supporting arms located on the frame, movable relative to the frame along a direction X; and

at least one press component located on the frame, wherein

each of the at least one press component comprises:

a first ball located on a side of a supporting arm of the at least two supporting arms away from another supporting arm of the at least two supporting arms, and the first ball configured to push the supporting arm elastically along a direction Y, wherein

the direction Y is substantially perpendicular to the direction X, the at least one press component is arranged along the direction X on the side of each of the at least two supporting arms.

**10**. The automated guided vehicle of claim **9**, wherein: the positioning and carrying structure further comprises a drive mechanism, the drive mechanism comprising a motor, a transmission assembly and a moving arm, the transmission assembly is connected to the motor and the moving arm, the moving arm is located on the frame and movable relative to the frame along the direction X, the moving arm is connected to the at least two supporting arms, the at least two supporting arms move with the moving arm.

**11**. The automated guided vehicle of claim **10**, wherein:

a gap is defined between the moving arm and the each of the at least two supporting arms in the direction X, the each of the at least two supporting arms is configured to change a position relative to the moving arm;

an adjusting bolt is connected between the moving arm and the each of the at least two supporting arms to fix a position of the each of the at least two supporting arms relative to the moving arm.

**12**. The automated guided vehicle of claim **9**, wherein: the positioning and carrying structure further comprises a positioning plate, the positioning plate is located on the each of the at least two supporting arms, the positioning plate is provided with a positioning column, the positioning column is configured for positioning an object to be transported.

**13**. The automated guided vehicle of claim **9**, wherein:

each of the support components comprises:

a second ball located below the each of the at least two supporting arms to support the each of the at least two supporting arms.

**14**. The automated guided vehicle of claim **9**, wherein:

each of the at least one press component comprises:

a shell;

an elastic member accommodated in the shell;

a guidepost accommodated in the shell;

one end of the elastic member rests against the inner wall of the shell, and another end of the elastic member bears against the guidepost, an end of the guidepost away from the elastic member is connected to the first ball.

**15**. The automated guided vehicle of claim **14**, wherein:

the frame comprises at least one groove, the shell is inserted in one groove of the at least one groove, a corresponding press component of the at least one press component is configured to change a position relative to the frame by adjusting the shell within the groove.

**16**. The automated guided vehicle of claim **9**, wherein:

the at least one press component is arranged along the direction X on the side of each of the at least two supporting arms.

**17**. The automated guided vehicle of claim **9**, wherein;

the positioning and carrying structure further comprises:

a limit switch located on the frame between the at least two supporting arms.

**18**. The automated guided vehicle of claim **9**, wherein:

the car comprises a lift configured to move the positioning and carrying structure up or down.

**19**. A loading system comprising:

an automated guided vehicle;

a fixture comprising a positioning groove; wherein

the automated guided vehicle comprises a positioning and carrying structure, wherein

the positioning and carrying structure comprises:

a frame;

at least two supporting arms located on the frame, movable relative to the frame along a direction X;

a positioning plate located on each of the at least two supporting arms, the positioning plate comprising a positioning column, the positioning column is insertable into the positioning groove;

at least one press component located on the frame, wherein

each of the at least one press component comprises:

a first ball located on a side of a supporting arm of the at least two supporting arms away from another supporting arm of the at least two supporting arms, and the first ball configured to push the supporting arm elastically along a direction Y;

the direction Y is substantially perpendicular to the direction X.

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